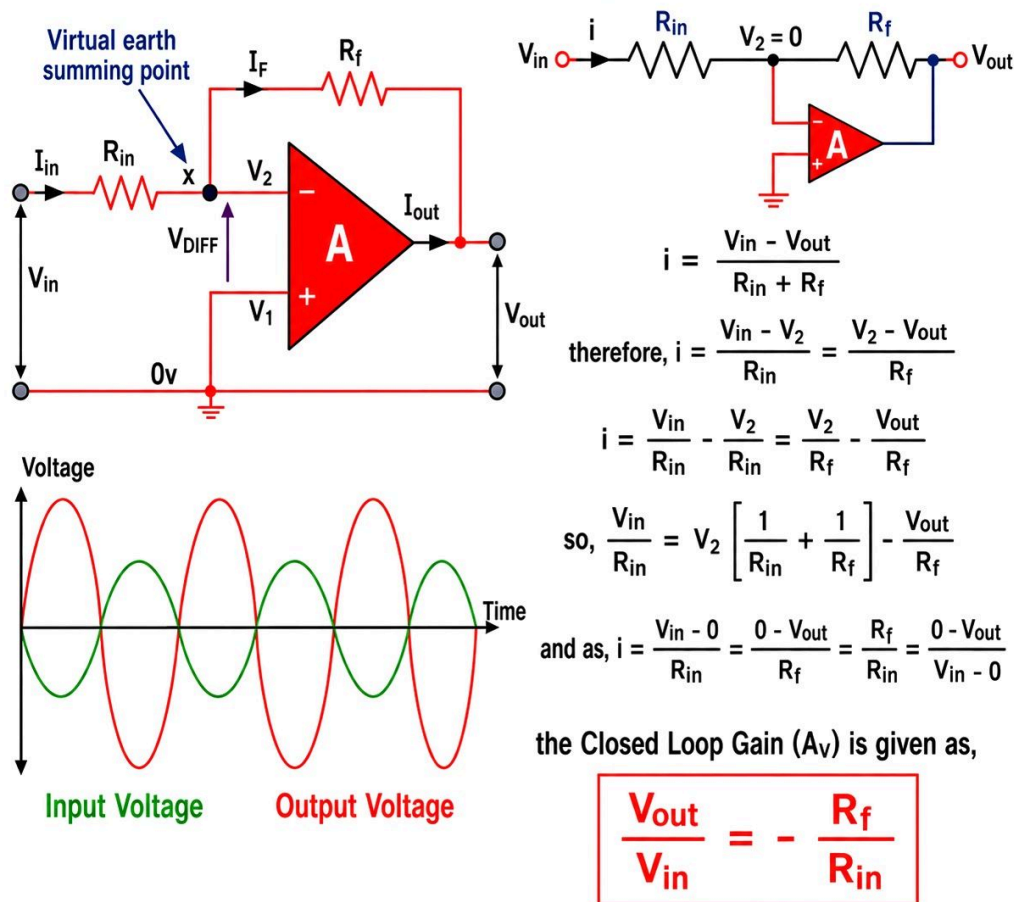




# OPAMP - INVERTING AMPLIFIER



Electronics Education

6 maj kl. 11:25 · 🌐

An inverting amplifier is one of the most widely used operational amplifier circuits in electronics. In this configuration, the input signal is applied to the inverting terminal through the resistor  $R_{in}$ , while the non-inverting terminal is connected to ground. Due to the very high gain of the op-amp and the negative feedback path through resistor  $R_f$ , the inverting input remains at nearly 0 volts. This point is called a virtual ground or virtual earth because it behaves like ground even though it is not directly connected to it.

When an input voltage is applied, current flows through the input resistor  $R_{in}$  toward the summing point. Since the op-amp input draws almost no current, the same current flows through the feedback resistor  $R_f$ . The op-amp continuously adjusts its output voltage to maintain the virtual ground condition. Because of this feedback action, the output signal becomes an amplified version of the input signal with opposite polarity. This means the output waveform is inverted by 180 degrees compared to the input waveform.

The voltage gain of the circuit depends on the ratio of the feedback resistor to the input resistor. The gain equation is:

$$V_{out} / V_{in} = -R_f / R_{in}$$

The negative sign shows that the output is inverted. If the feedback resistor is larger than the input resistor, the output amplitude increases. For example, if  $R_f$  is 10k ohms and  $R_{in}$  is 1k ohm, the gain becomes -10. This means the output voltage is ten times larger than the input but inverted.

Inverting amplifiers are commonly used in audio systems, active filters, signal processing circuits, instrumentation systems, and sensor interfaces because they provide stable gain, high accuracy, and

reliable performance.

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David Arthur Kuhnash

I've used an OPAMP utilizing a capacitive feedback to amplify the charge on a quartz crystal under pressure

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